SIMATS ENGINEERING
ASSESSMENT TOOL 1: ClassTest

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 10

Code of Conduct
I T . Kodanda Ram (192524162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Kodanda Ram

Assessment Tasks

ClassTest

1.Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2.Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4.Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5.Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

6.Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

7.Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

CSA 1518 - cloud computing

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CO4 - AT 1 - Class Test

(1) - A unified big data architecture can integrate customer data from both online & offline sources into a common analytics platform.

- Architecture

Unified Big Data Architecture for personalized Retail Analytics

online channels + offline channels

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Data ingestion

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Big data storage

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Data processing

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Real time analytics

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Recommendation Engine

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Bussiness applications

- It provides a unified customer view, enables real-time analysis, & gives personalized product recommendations.

28/30

(2) Distributed computing for scientific data processing.

- A distributed computing uses multiple connected computers to process large-scale scientific data instead of using a single centralized system.
- parallel processing divides a large task into smaller tasks and executes them simultaneously on different nodes. This reduces processing time, improves computational efficiency, and allows the system to handle increasing amounts of data.
- compared with centralized systems, distributed computing provides better scalability, faster processing & efficient resource utilization.

③ Big Data Fraud Detection Framework.

- A Fraud detection framework uses big data analytics to detect & prevent fraudulent transactions in digital payment platforms.
- streaming data processing continuously monitors transactions in real time & identifies suspicious activities quickly. predictive models analyze transaction history, customer behaviour, location, & payment patterns to calculate fraud risk.
- This helps improve detection accuracy, response time, & fraud prevention. It can also reduce financial losses by allowing suspicious transactions to be blocked or verified quickly.

④ Multimedia Data in Big Data.

- Multimedia data such as videos, audio & images in large in size & requires efficient storage & processing.
- The main challenges are high storage requirements, large data volume, & complex processing.
- Technologies such as HDFS, cloud storage, and distributed processing frameworks can be used to manage this data efficiently.
- These technologies provide scalability, faster processing, & efficient storage for large multimedia datasets.

⑤ Disaster Recovery & Bussiness continuity

- A disaster recovery strategy protects a cloud-based big data platform from failures & data loss.
- Redundancy keeps backup resources, while replication maintains copies of data at different locations.
- Automated failover switches to a backup system when

the primary system fails.
- These mechanisms ensure data availability, business continuity, and operational resilience with minimal downtime.

(6) Data quality in Big data analytics.
- Data quality issues such as inconsistency, duplication, & missing values can produce inaccurate results and reduce the reliability of analytics.
- Data governance can improve quality through data validation, deduplication, standardization, and proper data management policies.
- This ensures accurate, consistent, & reliable data for large scale analytics.

(7) Big data Recommendation system
- A recommendation system uses big data to suggest relevant content to users of an online streaming service.
- user behaviour data such as viewing history, ratings, & search patterns in analyzed to understand user preferences.
- Distributed processing handles large amounts of user data quickly & helps generate accurate recommendations.
- This improves recommendation accuracy & user engagement.

(8) Role of cloud computing in Big data
- cloud computing provides scalable & flexible resources for storing & processing big data.
- Its advantages includes scalability, flexibility and reduced infrastructure costs. However, challenges include cost management & security risks.

- cloud platforms allow organizations to increase resources when data grows & reduce them when demand decreases, making big data processing more efficient.

(9) Real time analytics for Transportation & logistics.

- A real-time analytics framework uses big data from vehicles, GPS, traffic, & delivery systems to improve transportation & logistics.
- Big data technologies analyze this information in realtime to support route optimization, fuel efficiency, & faster decision making. This helps identify traffic conditions & select better routes.

(10) Ethical & Legal challenges in Big data

- Big data collection can create privacy, security, & legal compliance issues, organizations must ensure that user data is collected & used responsibly.
- They should implement data privacy policies, access controls, data security, & regulatory compliance to protect users.
- This improves responsible data usage & user trust.